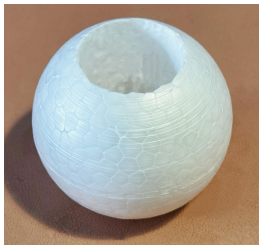
Created by Yuzhe (Liz) Liu and Sydney Pinkalla

Foam Bead Model



A 3-dimensional model of a bead (ball with a cylindrical hole) made out of expanded polystyrene foam (EPS, a.k.a. styrofoam) that can be used in various education settings. This bead model was used in a second year university-level Multivariable Calculus class to visualize setting up triple integrals under cylindrical and spherical coordinates.

Materials and Costs:

- **Styrofoam spheres**: Selectum - Styrofoam Balls - 2.83" Diameter - 4 Pack
    - $4.49 CAD per pack of four when purchased through Gwartzman's Art Supplies
- **Hot knife (Foam cutter)**: FloraCraft® Heated Tool CleanKut™ Foam Cutter
    - $29.99 CAD per cutter when purchased through Michaels
- **Sandpaper**: ~$5 CAD per paper
- **Pastry piping tip**: ~$3 CAD. We used one with a diameter of 1.5 inches.

Process of Prototyping:

- Took several tries to find the best tool (between typical knife, fruit corer, and foam cutter) and best cylinder size (2 inches versus 1.5 inches in diameter). In the end, only the foam cutter was able to create a smooth cut, to not break the foam sphere, and to not shed. The two cylinder sizes both worked, yet the 1.5" diameter (2:1 ratio with the foam sphere) was chosen as it better demonstrated the bead model under Multivariable Calculus learning goals.
- We also purchased many different brands of styrofoam, where the options purchased through Michaels shed too much and too easily (as they are only intended for use in flowery designs), where the option we ended up choosing was much cheaper, more durable when cutting and playing around with, and did not shed.

Step by Step:

1. Using the piping tip, create a dent on the styrofoam sphere on both sides for the potential area to cut out.
2. In a well-ventilated environment, charge and heat up the foam cutter.
3. After the cutter is heated, carefully insert it in the foam ball, following the dent created to cut out the cylinder shape. Repeat this on both sides.
4. Uncharge and safely place the cutter once cutting is complete. Then, push out the middle cylinder piece that is to be removed.
5. If uneven, charge the cutter again and very carefully cut off areas sticking out.
6. Using a sandpaper or other sanding tools, even out the middle cylinder hole.

Tips and Tricks:

- When cutting, wear essential protection gears (gloves, protection goggles/glasses, masks, etc.) and make sure the area is well-ventilated before proceeding.
- The foam cutter, once heated properly, makes cuts very rapidly and can cut over the line (i.e., makes larger cuts then intended). Hence, be very careful, cut slowly at the beginning, and cut smaller than the dent to ensure that the cut is in proportion.
- If possible, find securing tools to lock the foam spheres before cutting so it is easier to cut and make a perfect cut.

Photos:



Pack of models made for the Multivariable Calculus lesson.



Pack of middle cylinders removed when making the cylindrical holes.



Comparison of using a styrofoam cutter (on the left) versus using a typical knife (on the right).